

Crash course: how to make your research Bayesian

Syllabus

(1) Bayesian thinking and building intuition

- Probability and uncertainty
- Bayes' theorem as a way of updating beliefs
- Prior, likelihood, posterior, evidence
- Discrete examples

(2) Building simple inference models

- Continuous probabilities and probability density functions
- Calculating and interpreting 1D posteriors
- Prior dependence and sensitivity to assumptions

(3) More dimensions, comparing models

- Joint posteriors in multiple dimensions
- Marginalization over nuisance parameters
- Summarizing posteriors: credible intervals, contour and corner plots
- Predictions from the posterior
- Comparing competing models with the Bayes factor

(4) Bayesian renaissance: MCMC

- Why direct posterior evaluation fails in many dimensions
- Markov chain Monte Carlo and the Metropolis–Hastings algorithm
- A realistic inference example: foxes and rabbits, sampled with an ensemble sampler (**emcee**)
- Checking a chain: trace plots and autocorrelation